COS326 Examination November 2021

Test Information

Description

Examiner: Dr. Patricia E.N. Lutu **External examiner:** Prof. Roelien Goede

Examination date and time: 30 November 2021, 16:00 to 18:30

Time limit: 2 hours 30 minutes **Total marks:** 80 marks

1. This examination paper consists of 30 multiple choice questions of 1 mark each, 10 multiple choice questions of 3 marks each, and 4 short answer questions of 20 marks in total.
2. You may not consult any material during the examination.
3. In order to ask questions during the examination, you should click on the 'Ask questions during the exam' link on the ClickUP main menu.
4. The submission of this examination is subject to the University of Pretoria Integrity statement.

Integrity statement

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

Instructions	1. Attempt all questions 2. For technical reasons, the dollar sign is represented by the word 'dollar' in the questions.
Timed Test	This test has a time limit of 2 hours and 30 minutes.This test will save and be submitted automatically when the time expires. Warnings appear when **half the time**, **5 minutes**, **1 minute**, and **30 seconds** remain.*[The timer does not appear when previewing this test]*
Multiple Attempts	This Test allows multiple attempts.
Force Completion	This Test can be saved and resumed at any point until time has expired. The timer will continue to run if you leave the test.
	Your answers are saved automatically.

Question Completion Status:

QUESTION 1

1. State whether the following statement about the MongoDB document database is TRUE or FALSE:

   For the MongoDB document database, the primary construct used in the specification of queries and returning query results is the JSON object.

True

False

1 points

QUESTION 2

1. State whether the following statement is TRUE or FALSE:

   Four categories of NoSQL databases are: key-value stores, document databases, wide-column stores, and graph databases.

True

False

1 points

QUESTION 3

1. Three of the supporting technologies for the New SQL database management approach are:
   1. column-oriented data storage
   2. in-memory processing
   3. massively parallel processing

   Provide a brief explanation of how each of the above technologies supports the New SQL database management approach.

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6 points

QUESTION 4

1. Given that the following Cypher query has been executed on a Neo4j graph database:

```
CREATE
      ( psT:Person { name: "Thandi", from: "Durban",  hobby: "singing" }  ),
       ( psJ:Person { name: "Johan", from: "Pretoria",  hobby: "surfing" }   ),
       (psN:Person { name: "Neo", from: "Tshwane", hobby: "soccer"   }  ),

       (psT)-[:FOLLOWS { since: 2017 } ]->(psJ ),
       (psT)-[:FOLLOWS { since: 2016 } ]->(psN ),
       (psJ)-[:FOLLOWS  { since: 2018 } ]->(psT )
        (psJ)-[:FOLLOWS  { since: 2018 } ]->(psN )
```

   Which one of the following queries will display the details of the persons that Johan follows?

A.
```
MATCH    (psT {name: "Johan"}  )-[:FOLLOWS]->  (followed)
RETURN   name, from,  hobby;
```

B.
```
MATCH    (psT {name: "Johan"}  )-[:FOLLOWS]->  (followed)
RETURN   followed.name, followed.from,  followed.hobby;
```

C.
```
MATCH    (psT {name =  "Johan"}  )-[:FOLLOWS]->  (followed)
RETURN   psT.name, psT.from,  psT.hobby;
```

3 points

QUESTION 5

1. Given the following Java program declarations for a db4o object oriented database:

```
final static String DB4OFILENAME = System.getProperty("user.home") +
"/Driver.db4o";
ObjectContainer db =
        Db4oEmbedded.openFile(Db4oEmbedded.newConfiguration(), DB4OFILEN
AME);
```

Which one of the following pieces of Java code will perform an update operation?

A.
```
ObjectSet result =  db.queryByExample(new Driver ("Lewis Hamilton", 0));
Driver found = (Driver) result.next();
found.addPoints(10);
db.store(Driver);
```

B.
```
ObjectSet result =  db.queryByExample(new Driver ("Lewis Hamilton", 0));
Driver found = (Driver) result.next();
found.addPoints(10);
db.store(found);
```

C.
```
ObjectSet result =  db.query(new Driver ("Lewis Hamilton", 0));
Driver found = (Driver) result.next();
found.addPoints(10);
db.store(found);
```

3 points

QUESTION 6

1. Given the following contents of the collection *blogsCollection* in the MongoDB database *mydb*:

```
{
        _id:          ObjectId(7df78ad8902c)
        title:         'Cassandra Overview',
        description:    'Cassandra is a NoSql database',
        tags:          ['Cassandra', 'database', 'NoSQL'],
        likes:         100
},
{
        _id:          ObjectId(7df78ad8902d)
        title:         'NoSQL Overview',
        description:    'NoSql database is very fast',
        tags:          ['mongodb', 'neo4j', 'database', 'NoSQL'],
        likes:         200
},
{
        _id:          ObjectId(7df78ad8902e)
        title:         'VoltDB Overview',
        description:    'VoltDB is a New sql database',
        tags:          ['VoltDB', 'database', 'New SQL'],
        likes:         750
}
```

It is now required to write the JavaScript statement for the *mongo* shell to compute and display the total number of *likes* for all documents in the collection *blogsCollection* with the title 'NoSQL Overview'.

Which of the statements below will perform the required task?

A.
```
use mydb
db. blogsCollection.aggregate(
        [
                { dollar match: {title:"NoSQL Overview"} },
                  { dollar group:
                      {_id: "dollar title", num_likes: { dollar sum :
"dollar likes"}                    }
                }
        ] )
```

B.
```
use mydb
db. blogsCollection.aggregate(
        [
            { dollar match: {title:"NoSQL Overview"} },
            { dollar group by:
                  {_id: "dollar title", num_likes: { dollar sum : "dollar likes"} }
            }
        ] )
```

C.
```
use mydb
db. blogsCollection.aggregate(
        [
            { dollar match: {title:"NoSQL Overview"} },
            { dollar group:
                  {_id: "dollar title", num_likes: { dollar add : "dollar likes"} }
             }
        ] )
```

3 points

QUESTION 7

1. State whether the following statement about graph databases is TRUE or FALSE:

   A graph is represented as an object-oriented network of nodes (objects), edges (objects for node relationships) and properties which represent object attributes expressed as *key-value* pairings.

   True

   False

1 points

QUESTION 8

1. State whether the following statement about the MapReduce computational framework is TRUE or FALSE:

   There are two phases in a map-reduce computation. These are: map and reduce.

   True

   False

1 points

QUESTION 9

1. State whether the following statement about the MapReduce computational framework is TRUE or FALSE:

The MapReduce computational framework enables the implementation of highly distributed programs that run on failure-tolerant and scalable clusters of commodity machines.

True

False

1 points

QUESTION 10

1. State whether the following statement about New SQL database management systems is TRUE or FALSE:

   New SQL DBMSes support the relational data model and SQL as the primary mechanism for application interaction.

True

False

1 points

QUESTION 11

1. State whether the following statement about the data mining is TRUE or FALSE:

   Sentiment analysis of microblogs (e.g. tweets) can be conducted by storing the data in a NoSQL database and using text mining methods to determine the sentiment for each microblog.

True

False

1 points

QUESTION 12

1. State whether the following statement about New SQL database management systems is TRUE or FALSE:

   New SQL DBMSes aim to have the same level of scalability and availability as NoSQL DBMSes and maintain the ACID properties.

True

False

1 points

QUESTION 13

1. State whether the following statement about Object Oriented Database Management Systems is TRUE or FALSE:

   The class methods may be stored in the database and then the libraries are dynamically bound to the application at run-time.

True

False

1 points

QUESTION 14

1. State whether the following statement about graph databases is TRUE or FALSE:

The data in a graph database is stored using an adjacency matrix where each cell in the matrix stores a *key-value* pair for the relationship between two graph nodes.

True

False

1 points

QUESTION 15

1. Explain the similarities and differences between the data models for NoSQL databases which are classified as **key-value stores** and NoSQL databases which are classified as **wide-column stores**.

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6 points

QUESTION 16

1. State whether the following statement is TRUE or FALSE:

   For a key-value store, data searches can be performed on keys or values.

   True

   False

1 points

QUESTION 17

1. State whether the following statement about the MongoDB document database is TRUE or FALSE:

   Three types of queries for aggregation are: pipeline, map-reduce, and single purpose aggregation queries.

   True

   False

1 points

QUESTION 18

1. State whether the following statement about Object-relational databases (ORDBs) is TRUE or FALSE:

   ORDBs provide a seamless integration between applications and the database.

   True

   False

1 points

QUESTION 19

1. State whether the following statement about NoSQL database systems is TRUE or FALSE:

All NoSQL databases use the Big Data Model (BDM).

True

False

1 points

QUESTION 20

1. State whether the following statement about Object Oriented Database Management Systems is TRUE or FALSE:

   Objects may be organized into groups and one special object is responsible for storing the class methods on behalf of all the other objects in the group.

True

False

1 points

QUESTION 21

1. State whether the following statement is TRUE or FALSE:

   New SQL database management systems provide real-time OLTP and conventional OLAP in Big Data environments.

True

False

1 points

QUESTION 22

1. Suppose the following SQL statements have been executed on a PostgreSQL database:

```
CREATE SEQUENCE venueSeq START 101;
CREATE TYPE BuildingCodeType AS ENUM ('IT', 'LAW', 'CHM', 'PHY');
CREATE TYPE RoomCodeType AS (building BuildingCodeType, floor int, room int);
CREATE TYPE EquipmentType AS ENUM ('projector', 'PAsystem', 'safe', 'PC',
'phone');

CREATE TABLE Venue  (
                venueKey     integer DEFAULT nextval('venSeq') PRIMARY KEY,
                   venueCode    RoomCodeType,
                   seats  int  );

CREATE TABLE LectureRoom  (
                   equipmentList  text [ ] )
                   INHERITS (VENUE);
```

   It is now required to insert the following data into the LectureRoom table.
   venue key: auto-generated
   venue code: IT 4-2
   seats: 120
   equipment: projector, PAsystem, safe

   Which one of the following SQL statements will correctly insert the data into the LectureRoom table?

A. INSERT INTO LectureRoom (venueCode, seats, equipmentList)
VALUES (('IT', 4,2), 120, ARRAY['projector', 'PAsystem', 'safe']);

B. INSERT INTO LectureRoom (venueCode, seats, equipmentList)
VALUES (('IT', 4,2), 120, ['projector', 'PAsystem', 'safe']);

C. INSERT INTO LectureRoom (venueCode, seats, equipmentList)
VALUES (('IT', 4,2), 120, ('projector', 'PAsystem', 'safe'));

3 points

QUESTION 23

1. State whether the following statement about NoSQL database systems is TRUE or FALSE:

   All NoSQL databases provide Atomicity, Consistency, Isolation, Durability (ACID)
   guarantees for database transactions.

   True

   False

1 points

QUESTION 24

1. State whether the following statement is TRUE or FALSE:

   A NoSQL database is a distributed non-relational database system designed for large-scale data storage and for massively parallel data processing using powerful server machines.

   True

   False

1 points

QUESTION 25

1. Suppose the following SQL statements have been executed on a PostgreSQL database:

```
CREATE TABLE cities ( name text, population real, area int );
CREATE FUNCTION check_valid_area() RETURNS TRIGGER AS
dollar dollar
    BEGIN
        IF (NEW.area <= 0) THEN
            RAISE EXCEPTION 'Invalid area value!';
        END IF;
        RETURN NEW;
    END;
dollar dollar LANGUAGE plpgsql;
```

Which one of the following statements will create a trigger for the cities table?

A.
```
CREATE TRIGGER valid_area
    BEFORE INSERT OR UPDATE ON Cities
        FOR EACH ROW
            EXECUTE PROCEDURE check_valid_area();
```

B.
```
CREATE TRIGGER valid_area
    BEFORE INSERT OR UPDATE ON Cities
            EXECUTE PROCEDURE check_valid_area();
```

C.
```
CREATE TRIGGER valid_area
        FOR EACH ROW
```

```
EXECUTE PROCEDURE check_valid_area();
```

3 points

QUESTION 26

1. State whether the following statement about Object-relational databases (ORDBs) is TRUE or FALSE:

   ORDBs provide the object-oriented extensions to SQL, to database users. These
   extensions enable database users to implement user-defined types and inheritance.

True

False

1 points

QUESTION 27

1. State whether the following statement about graph databases is TRUE or FALSE:

   Three examples of application domains for graph databases are: representing social networks, generating recommendations, and conducting forensic investigations.

True

False

1 points

QUESTION 28

1. The manager of a business division wants an answer to the following question:

   "How much revenue did the gysmo product generate by province and by month this year?"

   Which one of the following SQL queries will produce the required information?

A.
```
SELECT B.province, T.month, SUM(S.revenue) AS gysmoRevenue
FROM BRANCH B, TIME T, PRODUCT P, SALES S
WHERE P.ProductName = 'gysmo'      AND
      T.year = 2021                AND
      B.BranchKey = S.BranchKey    AND
      T.TimeKey   = S.TimeKey      AND
      P.ProductKey = S.ProductKey
GROUP ON CUBE (B.province, T.month);
```

B.
```
SELECT B.province, T.month, SUM(S.revenue) AS gysmoRevenue
FROM BRANCH B, TIME T, PRODUCT P, SALES S
WHERE P.ProductName = 'gysmo' AND
      T.year = 2021                AND
      B.BranchKey = S.BranchKey    AND
      T.TimeKey   = S.TimeKey      AND
      P.ProductKey = S.ProductKey
GROUP BY ROLLUP (B.province, T.month);
```

C.
```
SELECT B.province, T.month, SUM(S.revenue) AS gysmoRevenue
FROM BRANCH B, TIME T, PRODUCT P, SALES S
WHERE P.ProductName = 'gysmo' AND
      T.year = 2021                AND
      B.BranchKey = S.BranchKey    AND
```

```
        T.TimeKey    = S.TimeKey          AND
        P.ProductKey = S.ProductKey
GROUP BY FIELDS (B.province, T.month);
```

3 points

QUESTION 29

1. State whether the following statement is TRUE or FALSE:

   Three characteristics of Big Data are big volume, big velocity and big verification.

   True

   False

1 points

QUESTION 30

1. State whether the following statement about the MapReduce computational framework is TRUE or FALSE:

   The MapReduce computational framework is one of the solutions for conducting fast computations in Big Data environments.

   True

   False

1 points

QUESTION 31

1. State whether the following statement about the data mining is TRUE or FALSE:

   Three modeling techniques that are commonly used data mining are: descriptive modeling, predictive modeling, and prescriptive modeling.

   True

   False

1 points

QUESTION 32

1. State whether the following statement about NoSQL database systems is TRUE or FALSE:

   Some NoSQL databases provide Atomicity, Consistency, Isolation, Durability (ACID) guarantees for database transactions while others only provide guarantees of eventual consistency.

   True

   False

1 points

QUESTION 33

1. Suppose the following SQL statements have been executed on a PostgreSQL database:

```
CREATE SEQUENCE venueSeq START 101;
CREATE TYPE BuildingCodeType AS ENUM ('IT', 'LAW', 'CHM', 'PHY');
CREATE TYPE RoomCodeType AS (building BuildingCodeType, floor int, room int);
```

```
CREATE TYPE EquipmentType AS ENUM ('projector', 'PAsystem', 'safe', 'PC',
'phone');

CREATE TABLE Venue  (
                    venueKey     integer DEFAULT nextval('venSeq') PRIMARY KEY,
                    venueCode    RoomCodeType,
                    seats  int  );

CREATE TABLE LectureRoom  (
                    equipmentList  text [ ] )
                    INHERITS (VENUE);
```

It is now required to implement the function *hasProjector*, which will return TRUE if a lecture room has a projector or FALSE otherwise. Which one of the following is a correct implementation of the *hasProjector* function?

A.
```
CREATE FUNCTION hasProjector ( text [ ]) RETURNS boolean AS
dollar dollar
    DELARE val text; proj boolean;
    BEGIN
                proj = FALSE;
                FOR val IN ARRAY dollar 1
                LOOP
                        IF (val = 'projector') THEN  proj = TRUE;   ENDIF;
                END LOOP
                RETURN  proj;
    END;
dollar dollar LANGUAGE plpgsql;
```

B.
```
CREATE FUNCTION hasProjector ( text [ ]) RETURNS boolean AS
dollar dollar
    DELARE val text; proj boolean;
    BEGIN
                proj = FALSE;
                FOREACH val IN ARRAY dollar 1
                LOOP
                        IF (val = 'projector') THEN  proj = TRUE;   ENDIF;
                END LOOP
                RETURN  proj;
    END;
dollar dollar LANGUAGE plpgsql;
```

C.
```
CREATE FUNCTION hasProjector ( text ) RETURNS boolean AS
dollar dollar
    DELARE val text; proj boolean;
    BEGIN
                proj = FALSE;
                FOREACH val IN ARRAY dollar 1
                LOOP
                        IF (val = 'projector') THEN  proj = TRUE;   ENDIF;
                END LOOP
                RETURN  proj;
    END;
dollar dollar LANGUAGE plpgsql;
```

3 points

QUESTION 34

1. One option for storing XML document data in a database is to store the XML document in a *shredded* form.

1. Give a concise description of this approach.
2. Discuss the strengths and weaknesses of this approach.

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4 points

QUESTION 35

1. State whether the following statement about the MongoDB document database is TRUE or FALSE:

   Two approaches to database design are: *embedded data model* and *de-normalized data model.*

   True

   False

1 points

QUESTION 36

1. Given that the following SQL statements have been executed on a PostgreSQL database:

   ```
   CREATE TYPE InventoryItem AS ( name text, supplierID char(5),
   unitPrice numeric);

   CREATE TABLE StockOnHand ( item inventoryItem, quantity integer);


   CREATE FUNCTION stockValue( inventoryItem, integer ) RETURNS numeric AS
   dollar dollar
       SELECT dollar 1.unitPrice * dollar 2 AS stockValue;
   dollar dollar LANGUAGE SQL;
   ```

   Which one of the following SELECT statements will produce a list of stock items on hand and the stock value, from the stockOnHand table?

A.
```
SELECT  item.name AS item_name,  item.supplierID AS supplier,
        item.unitPrice AS unit_price, quantity, stockValue(item, quantity)
FROM    stockOnHand;
```

B.
```
SELECT  (stockOnHand.item).name AS item_name,
        (stockOnHand.item).supplierID AS supplier,
        (stockOnHand.item).unitPrice AS unit_price,
        quantity, stockValue(item, quantity)
FROM    stockOnHand;
```

C.
```
SELECT  name  AS item_name, supplierID AS supplier,
        unitPrice  AS unit_price, quantity,
        stockValue(item, quantity)
FROM    stockOnHand;
```

3 points

QUESTION 37

1. State whether the following statement about Object-relational databases (ORDB) is TRUE or FALSE:

   ORDBs reduce the need for database table joins by enabling the storage of objects
   in tables.

True

False

1 points

QUESTION 38

1. The PostgreSQL database table below has been created using the following statement:
   CREATE TABLE XMLStaff (docNo CHAR(4), staffData XML);

The data shown below has then been imported into the table.

docNo	staffData
1	<STAFF branchNo = "B001"> <STAFFNO>S010</STAFFNO> <NAME> <FNAME>Anna</FNAME> <LNAME>Brown</LNAME> </NAME> </STAFF>
2	<STAFF branchNo = "B002"> <STAFFNO>S011</STAFFNO> <NAME> <FNAME>Thabiso</FNAME> <LNAME>Mashaba</LNAME> </NAME> </STAFF>
3	<STAFF branchNo = "B002"> <STAFFNO>S012</STAFFNO> <NAME> <FNAME>Jaco</FNAME> <LNAME>van Wyk</LNAME> </NAME> </STAFF>

Which one of the following PostgreSQL statements will display the staff number and name for all staff, in the following format?

StaffNumber	FirstName	Surname
S010	Anna	Brown
S011	Thabiso	Mashaba
S012	Jaco	van Wyk

A.
```
SELECT xpath('/STAFF/STAFFNO/text()', staffData) AS StaffNumber,
       xpath ('/STAFF/NAME/FNAME/text()', staffData) AS FirstName,
       xpath('/STAFF/NAME/LNAME/text()', staffData) AS Surname
FROM   XMLStaff;
```

B.
```
SELECT xpath('/STAFF/STAFFNO', staffData) AS StaffNumber,
       xpath('/STAFF/NAME/FNAME', staffData) AS FirstName,
       xpath('/STAFF/NAME/LNAME', staffData) AS Surname,
FROM   XMLStaff;
```

C.
```
SELECT unnest (xpath('/STAFF/STAFFNO/text()', staffData) ) AS StaffNumber,
       unnest (xpath('/STAFF/NAME/FNAME/text()', staffData) )
            AS FirstName,
       unnest (xpath('/STAFF/NAME/LNAME/text()', staffData) )
```

```
                    AS Surname,
FROM    XMLStaff;
```

3 points

QUESTION 39

1. Why are NoSQL databases suitable for the storage of Big Data?

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4 points

QUESTION 40

1. State whether the following statement about the data mining is TRUE or FALSE:

   Data mining is the process of extracting data from transactional databases, cleaning the data, and storing the clean data back into the database.

   True

   False

1 points

QUESTION 41

1. Given the following XML document:

```
<STAFFLIST>
<STAFF branchNo = "B005">
        <STAFFNO>SL21</STAFFNO>
        <NAME>
       <FNAME>John</FNAME><LNAME>Green</LNAME>
        </NAME>
        <SALARY>30000</SALARY>
</STAFF>
<STAFF branchNo = "B003">
        <STAFFNO>SG37</STAFFNO>
        <NAME>
       <FNAME>Ann</FNAME><LNAME>Brown</LNAME>
        </NAME>
        <SALARY>12000</SALARY>
</STAFF>
</STAFFLIST>
```

   Which one of the FLWOR queries below will list the staff names and staff numbers of all staff in ascending order of staff number?

A.

```
for dollar s in //STAFF
let dollar snum   :=   /STAFFNO
let dollar fn     :=   /NAME/FNAME
let dollar ln     :=   /NAME/LNAME
order by    /STAFFNO ascending
return      <STAFFDATA>
                    <STAFFNO>{ dollar snum/text() }</STAFFNO>
                    <FNAME>{ dollar fn/text() }</FNAME>
```

```
                <LNAME>{ dollar ln/text() }</LNAME>
            </STAFFDATA>
```

B.
```
let dollar snum   :=   /STAFFNO
let dollar fn     :=   /NAME/FNAME
let dollar ln     :=   /NAME/LNAME
order by     /STAFFNO ascending
return       dollar snum/text(), dollar fn/text(), dollar ln/text()
```

C.
```
for dollar s in //STAFF
let dollar snum   :=   dollar s/STAFFNO
let dollar fn     :=   dollar s/NAME/FNAME
let dollar ln     :=   dollar s/NAME/LNAME
order by dollar s/STAFFNO ascending
return       <STAFFDATA>
                <STAFFNO>{ dollar snum/text() }</STAFFNO>
                <FNAME>{ dollar fn/text() }</FNAME>
                <LNAME>{ dollar ln/text() }</LNAME>
            </STAFFDATA>
```

3 points

QUESTION 42

1. State whether the following statement about New SQL database management systems is TRUE or FALSE:

   New SQL DBMSes provide a brand new query language called New SQL.

True

False

1 points

QUESTION 43

1. State whether the following statement about Object Oriented Database Management Systems is TRUE or FALSE:

   The class methods may be stored in external files, and then the libraries are linked into the application at compile / link time.

True

False

1 points

QUESTION 44

1. State whether the following statement is TRUE or FALSE:

   Four categories of New SQL databases are: key-value stores, document databases, wide-column stores, and graph databases.

True

False

1 points

Click Save and Submit to save and submit. Click Save All Answers to save all answers.